



Fiona Stanley Fremantle Hospitals Group

OFFICIAL

Guideline

Medication Management of Agitation & Arousal in Mental Health Inpatients

Reference #:FSFH-MENH-GUI-0004

Scope

Site	Service/Department/Unit	Disciplines
Fiona Stanley Hospital Fremantle Hospital	Mental Health Service	Medical, Nursing, Allied Health, Pharmacy

1. Introduction

Safe and effective use of pro re nata (PRN) medications for pharmacological management of agitation and arousal in mental health inpatients depends on the appropriate selection of agents and monitoring of patients. This guideline supports the use of six different treatment algorithms based on the age and condition of the patient.

These guidelines apply to patients admitted to mental health inpatient units or other inpatient areas under the supervision of Consultation Liaison Psychiatry. They are not to be used for patients with delirium.

Refer to Management of Agitation in Older Adults with Dementia or Delirium (FSFH-HW-GUI-0006)

2. Terminology

Agitation and arousal

An acute mental state associated with an underlying mental and/or physical disorder in the form of: (i) agitation and distress, which is excessive verbal or motor activity that may or may not lead to aggression or violence; or (ii) actual aggression or violence entailing harm, hurt or injury to another person, or damage to property.

Antipsychotic	A medicine used to treat psychotic disorders. Antipsychotics are sometimes classified as typical (first generation) antipsychotics or atypical (second generation) antipsychotics. May also be used for a range of other mental health indications.
Benzodiazepine	These medicines are used to treat conditions such as anxiety, agitation, insomnia, alcohol withdrawal, muscle spasm and seizures.
PRN	'Pro re nata' is a Latin phrase used to indicate when medicines are to be used 'when required'. It is abbreviated to PRN.
TDS	Latin origin 'ter die sumendum' meaning three times a day
STAT	STAT is derived from the Latin word statim meaning immediately. A STAT medication order is charted on the front of the WA Hospital Medication Chart in the "Once Only, Pre-Medication and Nurse/Midwife Initiated Medications" section of the chart. They are only valid for 1 dose of the medication charted.

3. Guideline

Care should be taken with patients under 24 years, older adult patients (> 65 years) and patients who are:

- neuroleptic naïve
- benzodiazepine naïve
- respiratory compromised
- dehydrated and/or underweight
- intoxicated with drugs or alcohol
- medically compromised including cardiac disease, hepatic or renal impairment, epileptic
- pregnant
- sensitive to extrapyramidal side effects

These patients may respond to lower doses of medications, be more sensitive to side effects and require more frequent observation and monitoring.

3.1. Key Principles

- De-escalation or alternative coping strategies will be attempted in the first instance for mild to moderately aroused patients before progressing to psychotropic medications. All non-pharmacological interventions applied/attempted with patients should be documented

- Treatment is determined by patient's age and health status . Six algorithms (see appendices) map the treatment options for mental health patients in acute states of agitation or arousal.
- Medication for the management of agitation and arousal will be prescribed on the Agitation and Arousal PRN Medication Chart in mental health inpatient units
- The initial prescription of a Agitation and Arousal PRN Medication Chart must be reviewed by a Consultant Psychiatrist within 24 hours of the patients admission. Any changes to the medication regimen on the Agitation and Arousal PRN Medication Chart must be discussed with a Consultant Psychiatrist.
- Oral administration is preferred to parenteral administration wherever possible.
- Orally disintegrating tablets/wafers, liquids or syrups may be used in preference to tablets to aid compliance.
- Doses must be clearly defined when prescribed, not a dosage range.
- Maximum daily doses must be specified. The maximum dose / 24hrs is the total dose (via any route) allowable for that medication to be administered by any order inclusive of any regular, other PRN or STAT medication.
- Benztropine (oral or intramuscular) 1mg to 2mg 'TDS PRN' MUST be prescribed on the PRN section of the WA Hospital Medication Chart (WAHMC) for patients prescribed antipsychotics on the Agitation and Arousal PRN Medication Chart
- Flumazenil must be available if benzodiazepines are administered intramuscularly .
- Frequent PRN administrations (as determined by clinical judgement) for greater than 24 hours, repeated episodes of agitation and arousal and/or inadequate response to PRN should prompt a review of the patient's regular medication regimen.
- The Medical Officer on duty is to be informed when intramuscular medication is administered.
- PRN usage will be reviewed and documented in the digital medical record (DMR) by a Medical Officer every 24 hours for patients requiring agitation and arousal medication.
- The aim of pharmacological management is to reduce the patient's agitation and arousal score to 1, not to sedate the patient.
- Seek the advice of a Consultant Psychiatrist if not achieving a score of 1 at maximum doses recommended in the corresponding appendices.

- Consultant Psychiatrist approval is required for the use of Acuphase® for managing acute arousal and agitation. Patients must be assessed by a Consultant Psychiatrist or a registrar under direction of a Consultant Psychiatrist prior to each dose. Refer to Zuclopenthizol acetate (Clopixol Acuphase ®) Specialised Drug Guideline FSFH-PHARM-GUI-001.

3.2. Medication Selection Principles

- Midazolam for intramuscular injection is not available at Fiona Stanley Hospital – Cockburn Health (hence forward referred to as Cockburn Health) site. If an IM benzodiazepine is required for agitation and arousal management, use lorazepam.
- Benzodiazepines should be used with caution or avoided altogether in conditions that may predispose patients to respiratory compromise. These conditions may include:
 - Obstructive sleep apnoea
 - Acute exacerbation of severe airways disease (such as chronic obstructive pulmonary disease, asthma, occupational lung diseases and pulmonary hypertension)
 - Bronchiectasis
 - Lower respiratory tract infection (including patients who are COVID-19 positive OR who display COVID-19 symptoms OR are contacts of COVID-19)
 - Neuromuscular disorders (such as myasthenia gravis, motor neurone disease and amyotrophic lateral sclerosis)
 - See [Appendix 1](#) and [Appendix 2](#) for modified Agitation and Arousal Guidelines
- Benzodiazepines should be avoided in patients with dementia
 - There is limited evidence of benefit in this patient cohort and they are associated with serious adverse effects including cognitive decline, urinary incontinence and falls.
- Benzodiazepines should be avoided in patients with delirium unless delirium is related to alcohol or benzodiazepine withdrawal or seizures.
 - Benzodiazepines increases delirium risk and adverse effects, including falls, are common.
- If benzodiazepines are required those with a short half life (i.e lorazepam) should be used.
- Do not use haloperidol in established or suspected Parkinson's disease or Lewy Body Dementia

- In patients requiring treatment for Covid 19, ensure interactions with antiviral medications are checked using [Liverpool Covid 19 Interaction Checker](#)

3.3. Patient monitoring

- Continuous direct visual observation (not closed circuit television) must be maintained following intramuscular medication administration until it is clear that the patient is not over-sedated and is maintaining their airway.
- Following intramuscular administration of PRN medication, the risk of falls and orthostatic hypotension is increased and physiological observations will be recorded:
 - every 10 minutes for one hour,
 - then half hourly for a minimum of four hours or until the patient is ambulatory.
- All physical observations will be documented on the Adult Observation and Response Chart (AORC).
- If the patient refuses or it is unsafe to complete the observations this will be documented on the AORC chart and digital medical record.
- Agitation and arousal scores before and after each PRN dose will be documented.
- Monitor for extrapyramidal side effects for 48 hours following the administration of antipsychotics.

3.4. Agitation and arousal scores

- Agitation and arousal scores will be documented on the Agitation and Arousal PRN Medication Chart.

Route	Scores to be recorded at the following times:		
Oral	Prior to dose	60 minutes	90 minutes
Intramuscular	Prior to dose	30 minutes	60 minutes

Agitation and arousal rating scale		
Score	Observation	Management
0	Asleep or Unconscious	Review doses and follow monitoring requirements
1	Settled, minimal agitation, PRN not required	Optimal

2	Mildly aroused, pacing, still willing to talk reasonably	1. Verbal de-escalation as appropriate 2. Oral (PO) treatment option if willing to take oral medication 3. Intramuscular (IM) only if unwilling to take oral medication
3	Moderately aroused, agitated, becoming more vocal and unreasonable or hostile	
4	Highly aroused and possibly distressed or fearful, possibly noisy but not violent	
5	Highly aroused and violent toward self, others and/or property	

3.5. Repeat Doses

- For patients less than 65 years of age:
 - Oral doses may be repeated at hourly intervals unless otherwise specified
- For patients aged 65 years and above:
 - Oral doses may be repeated at 2 hourly intervals unless other wise specified.
- For all patients, intramuscular doses may be repeated after 20 minutes, unless other wise specified, except for olanzapine IM.
 - Olanzapine IM may be repeated after 2 hours and a third dose 6 hours after the first dose administered. Total of 3 doses or 20mg maximum in 24 hours whichever occurs first.
- If the prescriber wants an hourly frequency that differs from the standard hourly frequencies above then this must be clearly annotated in the medication order under the dose.
- If a repeat dose is thought to be necessary prior to the stipulated time interval, a medical review is required.

4. Compliance/Performance Monitoring

- The Pharmacy Department will be responsible for monitoring compliance with this document. Compliance will be monitored via routine clinical incident review processes.

5. Related Policy Documents

- WA Heath:
 - [Management of Acute Behavioural Disturbance in COVID-19](#)
 - [Guidelines for the use of the WA Agitation and Arousal PRN Medication Chart](#)
- KEMH:
 - [Management of Mental Health Emergency](#)
- PCH:
 - [Agitation and Arousal Medication Management](#)
- FSFHG:
 - [Management of Agitation in Older Adults with Dementia or Delirium](#)
 - [FSFHG Medication Administration](#)
 - [Zuclopenthizol acetate \(Clopixol Acuphase ®\) Specialised Drug Guideline](#)

6. Related Standards

- NSQHS Standards:
 - Medication Safety
 - Comprehensive Care
 - Communicating for Safety

7. Bibliography

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AusDI by Telstra Health. (2025). Product Information. Retrieved from: <https://ausdi-hcn-com-au.smhslibresources.health.wa.gov.au/quickSearch.hcn>

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FSFH-MENH-GUI-0004	Version 4.0	Endorsed July 2025	Review Due July 2028	Page 7 of 14
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The hard copy of this document may be out of date. To ensure you are reading the current version, check the policy and guideline site on the FSFHG Policy Intranet.

8. Authorisation

EXECUTIVE SPONSOR: Nurse Director, Service 5					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Endorsed By	Revision due
1	09/2018	FH Senior pharmacist FSH Senior pharmacist	DTC MHS Safety Quality Risk Committee	Policy Committee	09/2021
2	04/2022	FH Senior Pharmacist	DTC NMPGC	S5 SQR Committee	04/2025
3	11/2022	FSH Senior Pharmacist	DTC	S5 SQR Committee	01/2026
4.	07/2025	FH MH Team Lead Pharmacist	DTC, NMPGC	S5 SQR Committee	07/2028

9. Appendices

9.1. Appendix 1: Management of Adult (18-65 years) with a condition that predisposes to respiratory compromise

Appendix 1: Agitation and arousal pharmacological management for ADULT mental health inpatients from <u>18 to 65 YEARS</u> who have a condition that predisposes them to respiratory compromise. Refer to section 3.3 above for an overview of conditions that may predispose patients to respiratory compromise.																											
De-escalation techniques will be attempted first for mild to moderately aroused patients before progressing to psychotropic medications.																											
ORAL USE ORAL MEDICATION FIRST WHENEVER POSSIBLE			INTRAMUSCULAR *ECG HIGHLY RECOMMENDED																								
STEP 1 MILD TO MODERATE AROUSAL Agitation and arousal rating scale 2-3 and willing to take oral medication	STEP 2 MODERATE TO SEVERE AROUSAL Agitation and arousal rating scale 3-5 and willing to take oral medication.	STEP 3 MODERATE TO SEVERE AROUSAL Agitation and arousal rating scale 3-5 and willing to take oral medication.	STEP 4 MODERATE TO SEVERE AROUSAL Agitation and arousal rating scale 3-5 and UNWILLING to take oral medication																								
SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES STEP 1: Haloperidol PO 5mg OR STEP 1A: Quetiapine Immediate Release PO 25 to 50mg OR Step 1B: Olanzapine PO 2.5 to 5mg	SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES STEP 2: Haloperidol PO 10mg OR STEP 2A: Quetiapine Immediate Release PO 25 to 100mg OR Step 2B: Olanzapine PO 5 to 10mg	SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES STEP 3: Quetiapine Immediate Release PO 100 to 200mg OR STEP 3A: Olanzapine PO 5 to 10mg	SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES STEP 4: Droperidol IM 5 to 10mg OR STEP 4A: Olanzapine IM 5 to 10mg Do not administer olanzapine IM within 1 hour before or after any parenteral benzodiazepine																								
INFORMATION: MONITOR: - Observations including but not limited to blood pressure and respiratory function every 10 minutes for 1 hour, and then half hourly until the patient is ambulatory or at least 4 hours following IM drug administration. - QT prolongation, acute dystonia and neuroleptic malignant syndrome with antipsychotics. - ECG is HIGHLY RECOMMENDED prior to and following IM antipsychotic administration to assess for prolongation of QT interval. - Frequent PRN administrations for more than 24 hours should prompt review of regular medications.		REVIEW RESPONSE: - Review IM doses after 30 and 60 min and record arousal rating scale. - Review all oral doses after 60 or 90 min and record arousal rating scale. - Scores are recorded on Agitation and Arousal PRN Chart. REPEAT DOSING: - Where additional doses are required, administer at the frequency indicated until score of 1 or maximum daily dosage is achieved. - Oral: May be repeated hourly - IM: May be repeated after 20 minutes except for Olanzapine IM. Olanzapine IM may be repeated after 2 hours and a third dose 6 hours after the first dose administered. Total of 3 doses or 20mg maximum in 24 hours whichever occurs first. - Seek the advice of a psychiatrist if not achieving a score of 1 at maximum doses recommended																									
		PSYCHOTROPIC DRUG SELECTION AND DOSES: - Where possible use PRN medications that are the same or similar to regular medications. - Dosing frequencies outside of recommended time and/or dose range require psychiatrist approval. - Benzodiazepines are not appropriate in patients with conditions that predispose them to respiratory compromise - Olanzapine is preferred medication for patients prescribed Paxlovid® for treatment of Covid 19. Haloperidol and quetiapine metabolism is reduced by Paxlovid® OTHER MEDICATIONS: - Benzatropine (oral/IM) 1mg-2mg TDS PRN must always be prescribed on the PRN section of the Western Australia Hospital Medication Chart NOTIFICATION: - After an IM dose is administered, the doctor on duty will be notified.																									
		RECOMMENDED MAXIMUM DOSES: <table border="1"> <thead> <tr> <th>Medication</th><th>Recommended maximum PRN dose</th><th>single dose</th><th>Total maximum oral AND IM daily dose (including regular)</th></tr> </thead> <tbody> <tr> <td>Oral Haloperidol</td><td>10mg</td><td></td><td>30mg</td></tr> <tr> <td>Oral Olanzapine</td><td>10mg</td><td></td><td>30mg</td></tr> <tr> <td>Oral Quetiapine</td><td>200mg</td><td></td><td>1000mg</td></tr> <tr> <td>IM Droperidol</td><td>10mg</td><td></td><td>20mg</td></tr> <tr> <td>IM Olanzapine</td><td>10mg</td><td></td><td>20mg</td></tr> </tbody> </table>		Medication	Recommended maximum PRN dose	single dose	Total maximum oral AND IM daily dose (including regular)	Oral Haloperidol	10mg		30mg	Oral Olanzapine	10mg		30mg	Oral Quetiapine	200mg		1000mg	IM Droperidol	10mg		20mg	IM Olanzapine	10mg		20mg
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Please note: some antivirals used to treat COVID-19 may interact psychotropic medications. Seek specialist advice and use the [COVID-19 Drug Interaction](#) checker

9.2. Appendix 2 - Management of Older Adult (>65 years) with a condition that predisposes to respiratory compromise

Appendix 2: Agitation and arousal pharmacological management for OLDER ADULT mental health inpatients OVER 65 YEARS who have a condition that predisposes them to respiratory compromise

Refer to section 3.3 above for an overview of conditions that may predispose patients to respiratory compromise.

De-escalation techniques will be attempted first for mild to moderately aroused patients before progressing to psychotropic medications.

ORAL USE ORAL MEDICATION FIRST WHENEVER POSSIBLE			INTRAMUSCULAR *ECG HIGHLY RECOMMENDED																		
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9.3. Appendix 3: Mangement for Older Adolescents (16-17 years)

Appendix 3: Acute agitation and arousal pharmacological management for YOUTH mental health inpatients from <u>16 to 17 YEARS</u>																											
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Please note: some antivirals used to treat COVID-19 may interact psychotropic medications. Seek specialist advice and use the [COVID-19 Drug Interaction](#) Check

9.4. Appendix 4: Management for Adults (18-65 years)

Appendix 4: Agitation and arousal pharmacological management for ADULT mental health inpatients from 18 to 65 YEARS																														
De-escalation techniques will be attempted first for mild to moderately aroused patients before progressing to psychotropic medications. For patients with conditions that predispose them to respiratory compromise, see Appendix 1																														
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STEP 1: Lorazepam PO/ subling 1 to 2.5mg	STEP 2: Lorazepam PO/subling 1 to 2.5mg : STEP 2A: Quetiapine Immediate Release PO 25 to 100mg OR STEP 2B: Olanzapine PO 5 to 10mg OR STEP 2C: Chlorpromazine PO 25 to 100mg OR STEP 2D: Combinations of above antipsychotic AND Lorazepam PO/subling 1 to 2.5mg	STEP 3: Quetiapine Immediate Release PO 100 to 200mg OR STEP 3A: Olanzapine PO 5 to 10mg OR STEP 3B: Chlorpromazine PO 100 to 200mg OR STEP 3C: Combinations of above antipsychotic AND Lorazepam PO/subling 1 to 2.5mg	STEP 4: Lorazepam IM 2 to 4mg OR STEP 4A: Olanzapine IM 5 to 10mg OR Do not administer IM Olanzapine within 1 hour before or after any parenteral benzodiazepine STEP 4B: Midazolam IM 2.5 to 10mg Note: Midazolam is not used at Cockburn Health. If an IM benzodiazepine is required, use lorazepam OR STEP 4C: Droperidol IM 5 to 10mg OR STEP 4D: Lorazepam IM 2 to 4mg AND Droperidol IM 5mg OR STEP 4E: Midazolam IM 5 to 10mg) AND Droperidol IM 5mg Note: Midazolam is not used at Cockburn Health. If an IM benzodiazepine is required, use lorazepam.																											
INFORMATION: MONITOR: - Observations including but not limited to blood pressure and respiratory function every 10 minutes for 1 hour, and then half hourly until the patient is ambulatory or at least 4 hours following IM drug administration. - QT prolongation, acute dystonia and neuroleptic malignant syndrome with antipsychotics. - ECG is HIGHLY RECOMMENDED prior to and following IM antipsychotic administration to assess for prolongation of QT interval. - Frequent PRN administrations for more than 24 hours should prompt review of regular medications.		REVIEW RESPONSE: - Review IM doses after 30 and 60 min and record arousal rating scale. - Review all oral doses after 60 or 90 min and record arousal rating scale. - Scores are recorded on Agitation and Arousal PRN Chart. REPEAT DOSING: - Where additional doses are required, administer at the frequency indicated until score of 1 or maximum daily dosage is achieved. - Oral: May be repeated hourly - IM: May be repeated after 20 minutes except for Olanzapine IM. Olanzapine IM may be repeated after 2 hours and a third dose 6 hours after the first dose administered. Total of 3 doses or 20mg maximum in 24 hours whichever occurs first. - Seek the advice of a psychiatrist if not achieving a score of 1 at maximum doses recommended																												
		PSYCHOTROPIC DRUG SELECTION AND DOSES: - Where possible use PRN medications that are the same or similar to regular medications. - Dosing frequencies outside of recommended time and/or dose range require psychiatrist approval. OTHER MEDICATIONS: - Ensure Flumazenil is available if benzodiazepines are administered parenterally - Benzatropine (oral/IM) 1mg-2mg TDS PRN must always be prescribed on the PRN section of the Western Australia Hospital Medication Chart NOTIFICATION: - After an IM dose is administered, the doctor on duty will be notified.	RECOMMENDED MAXIMUM DOSES: <table><tr><th>Medication</th><th>Recommended maximum single PRN dose</th><th>Total Maximum Oral AND IM Daily Dose (including regular)</th></tr><tr><td>Oral Chlorpromazine</td><td>200mg</td><td>1000mg</td></tr><tr><td>Oral Lorazepam</td><td>2.5mg</td><td>10mg</td></tr><tr><td>Oral Olanzapine</td><td>10mg</td><td>30mg</td></tr><tr><td>Oral Quetiapine</td><td>200mg</td><td>1000mg</td></tr><tr><td>IM Droperidol</td><td>10mg</td><td>20mg</td></tr><tr><td>IM Lorazepam</td><td>4mg</td><td>10mg</td></tr><tr><td>IM Midazolam</td><td>10mg</td><td>20mg</td></tr><tr><td>IM Olanzapine</td><td>10mg</td><td>20mg</td></tr></table>	Medication	Recommended maximum single PRN dose	Total Maximum Oral AND IM Daily Dose (including regular)	Oral Chlorpromazine	200mg	1000mg	Oral Lorazepam	2.5mg	10mg	Oral Olanzapine	10mg	30mg	Oral Quetiapine	200mg	1000mg	IM Droperidol	10mg	20mg	IM Lorazepam	4mg	10mg	IM Midazolam	10mg	20mg	IM Olanzapine	10mg	20mg
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Oral Olanzapine	10mg	30mg																												
Oral Quetiapine	200mg	1000mg																												
IM Droperidol	10mg	20mg																												
IM Lorazepam	4mg	10mg																												
IM Midazolam	10mg	20mg																												
IM Olanzapine	10mg	20mg																												

9.5. Appendix 5: Management for Older Adults (>65 years)

Appendix 5: Acute agitation and arousal pharmacological management for OLDER ADULT mental health inpatients OVER 65 YEARS

De-escalation techniques will be attempted first for mild to moderately aroused patients before progressing to psychotropic medications.
For patients with conditions that predispose them to respiratory compromise, see Appendix 2

ORAL USE ORAL MEDICATION FIRST WHENEVER POSSIBLE			INTRAMUSCULAR *ECG HIGHLY RECOMMENDED																																
STEP 1 MILD TO MODERATE AROUSAL Agitation and arousal rating scale 2-3 and willing to take oral medication	STEP 2 MODERATE TO SEVERE AROUSAL Agitation and arousal rating scale 3-5 and willing to take oral medication.	STEP 3 MODERATE TO SEVERE AROUSAL Agitation and arousal rating scale 3-5 and willing to take oral medication.	STEP 4 MODERATE TO SEVERE AROUSAL Agitation and arousal rating scale 3-5 and <u>UNWILLING</u> to take oral medication																																
SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES	SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES	SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES	SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES																																
STEP 1 Haloperidol PO 0.25 to 0.5mg OR Step 1A Quetiapine Immediate Release PO 12.5 to 25mg **Oral benzodiazepines may be appropriate in certain patient populations. Must be discussed with a Consultant Psychiatrist prior to prescribing. **	STEP 2A: Quetiapine Immediate Release PO 25 to 50mg OR Step 2B: Haloperidol PO 0.25 to 0.5mg OR Step 2C: Olanzapine PO/subling 2.5mg **Oral benzodiazepines may be appropriate in certain patient populations. Must be discussed with a Consultant Psychiatrist prior to prescribing. **	STEP 3A: Quetiapine Immediate Release PO 50 mg OR STEP 3B: Haloperidol PO 0.5 to 1mg OR Step 3B: Olanzapine PO/sublin 2.5 to 5mg **Oral benzodiazepines may be appropriate in certain patient populations. Must be discussed with a Consultant Psychiatrist prior to prescribing. **	STEP 4A: Haloperidol IM 0.25 to 0.5mg OR Step 4B: Olanzapine IM 2.5 to 5mg Do not administer IM Olanzapine within 1 hour before or after any parenteral benzodiazepine **IM benzodiazepines may be appropriate in certain patient populations. Must be discussed with a Consultant Psychiatrist prior to prescribing.**																																
INFORMATION: MONITOR: - Observations including but not limited to blood pressure and respiratory function every 10 minutes for 1 hour, and then half hourly until the patient is ambulatory or at least 4 hours following IM drug administration. - QT prolongation, acute dystonia and neuroleptic malignant syndrome with antipsychotics. - ECG is HIGHLY RECOMMENDED prior to and following IM antipsychotic administration to assess for prolongation of QT interval. - Frequent PRN administrations for more than 24 hours should prompt review of regular medications.																																			
REVIEW RESPONSE: - Review IM doses after 30 and 60 min and record arousal rating scale. - Review all oral doses after 60 or 90 min and record arousal rating scale. - Scores are recorded on Agitation and Arousal PRN Chart. REPEAT DOSING: - Where additional doses are required, administer at the frequency indicated until score of 1 or maximum daily dosage is achieved. - Oral: May be repeated 2 hourly - IM: May be repeated after 20 minutes except for Olanzapine IM. Olanzapine IM may be repeated after 2 hours and a third dose 6 hours after the first dose administered. Total of 3 doses or 20mg maximum in 24 hours whichever occurs first. - Seek the advice of a psychiatrist if not achieving a score of 1 at maximum doses recommended																																			
PSYCHOTROPIC DRUG SELECTION AND DOSES: - Where possible use PRN medications that are the same or similar to regular medications. - Do not use haloperidol in established or suspected Parkinson's disease or Lewy Body Dementia - Dosing frequencies outside of recommended time and/or dose range require psychiatrist approval. - Lorazepam may be appropriate for patients without dementia or delirium. Consultant Psychiatrist approval required before prescribing. OTHER MEDICATIONS: - Ensure Flumazenil is available if benzodiazepines are administered parenterally - Benzatropine (oral/IM) 1mg-2mg TDS PRN must always be prescribed on the PRN section of the Western Australia Hospital Medication Chart NOTIFICATION: - After an IM dose is administered, the doctor on duty will be notified.																																			
RECOMMENDED MAXIMUM DOSES: <table border="1"> <thead> <tr> <th>Medication</th><th>Recommended maximum single dose</th><th>PRN</th><th>Total maximum oral AND IM daily dose (including regular)</th></tr> </thead> <tbody> <tr> <td>Oral Lorazepam</td><td>1mg</td><td></td><td>4mg</td></tr> <tr> <td>Oral Haloperidol</td><td>1mg</td><td></td><td>10mg</td></tr> <tr> <td>Oral Quetiapine</td><td>50mg</td><td></td><td>300mg</td></tr> <tr> <td>Oral Olanzapine</td><td>5mg</td><td></td><td>20mg</td></tr> <tr> <td>IM Lorazepam</td><td>1mg</td><td></td><td>4mg</td></tr> <tr> <td>IM Haloperidol</td><td>1mg</td><td></td><td>5mg</td></tr> <tr> <td>IM Olanzapine</td><td>5mg</td><td></td><td>20mg</td></tr> </tbody> </table>				Medication	Recommended maximum single dose	PRN	Total maximum oral AND IM daily dose (including regular)	Oral Lorazepam	1mg		4mg	Oral Haloperidol	1mg		10mg	Oral Quetiapine	50mg		300mg	Oral Olanzapine	5mg		20mg	IM Lorazepam	1mg		4mg	IM Haloperidol	1mg		5mg	IM Olanzapine	5mg		20mg
Medication	Recommended maximum single dose	PRN	Total maximum oral AND IM daily dose (including regular)																																
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IM Lorazepam	1mg		4mg																																
IM Haloperidol	1mg		5mg																																
IM Olanzapine	5mg		20mg																																

9.6. Appendix 6: Management of Pregnant Women

Appendix 6: Acute agitation and arousal pharmacological management for PREGNANT mental health inpatients aged 16 YEARS and above

De-escalation techniques will be attempted first for mild to moderately aroused patients before progressing to psychotropic medications.
For patients with conditions that predispose them to respiratory compromise, Seek specialist advice.

ORAL USE ORAL MEDICATION FIRST WHENEVER POSSIBLE			INTRAMUSCULAR *ECG HIGHLY RECOMMENDED																					
STEP 1 MILD TO MODERATE AROUSAL Agitation and arousal rating scale 2-3 and willing to take oral medication SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES STEP 1: Lorazepam PO/subling 0.5 to 1mg OR STEP 1A Quetiapine Immediate Release PO 12.5mg to 25mg OR STEP 1B: Olanzapine PO/subling 2.5mg	STEP 2 MODERATE TO SEVERE AROUSAL Agitation and arousal rating scale 3-5 and willing to take oral medication. SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES STEP 2: Lorazepam PO/subling 1 to 2mg OR STEP 2A: Quetiapine Immediate Release PO 25 to 50mg OR STEP 2B Olanzapine PO/subling 5mg	STEP 3 MODERATE TO SEVERE AROUSAL Agitation and arousal rating scale 3-5 and willing to take oral medication. SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES STEP 3: Quetiapine Immediate Release PO 25 to 50mg AND Lorazepam PO/subling 1 to 2mg	STEP 4 MODERATE TO SEVERE AROUSAL Agitation and arousal rating scale 3-5 and UNWILLING to take oral medication SPECIFY EXACT DOSE – DO NOT USE DOSING RANGES STEP 4: Midazolam IM 2.5 to 5mg (Note: Midazolam IM is not used at Cockburn Health. If an IM benzodiazepine is required, used lorazepam.= OR STEP 4A: Lorazepam IM 0.5 to 1mg OR STEP 4B Olanzapine IM 5mg Do not administer IM Olanzapine within 1 hour before or after any parenteral benzodiazepine Seek anaesthetics support																					
INFORMATION: MONITOR: - Observations including but not limited to blood pressure and respiratory function every 10 minutes for 1 hour, and then half hourly until the patient is ambulatory or at least 4 hours following IM drug administration. - QT prolongation, acute dystonia and neuroleptic malignant syndrome with antipsychotics. - ECG is HIGHLY RECOMMENDED prior to and following IM antipsychotic administration to assess for prolongation of QT interval. - Frequent PRN administrations for more than 24 hours should prompt review of regular medications.		REVIEW RESPONSE: - Review IM doses after 30 and 60 min and record arousal rating scale. - Review all oral doses after 60 or 90 min and record arousal rating scale. - Scores are recorded on Agitation and Arousal PRN Chart. REPEAT DOSING: - Where additional doses are required, administer at the frequency indicated until score of 1 or maximum daily dosage is achieved. - Oral: May be repeated hourly - IM: May be repeated after 20 minutes except for Olanzapine IM. Olanzapine IM may be repeated after 2 hours and a third dose 6 hours after the first dose administered. Total of 3 doses or 20mg maximum in 24 hours whichever occurs first. - Seek the advice of a psychiatrist if not achieving a score of 1 at maximum doses recommended																						
		PSYCHOTROPIC DRUG SELECTION AND DOSES: - Where possible use PRN medications that are the same or similar to regular medications. - Dosing frequencies outside of recommended time and/or dose range require psychiatrist approval. - Quetiapine is the only licensed medication for older adolescents in Australia. - Provision of off-label treatment to any involuntary child must be reported to the Chief Psychiatrist. OTHER MEDICATIONS: - Ensure Flumazenil is available if benzodiazepines are administered parenterally. Caution in benzodiazepine-dependent women; risk of precipitating withdrawal in fetus. - Benzatropine (oral/IM) 1mg-2mg TDS PRN must always be prescribed on the PRN section of the Western Australia Hospital Medication Chart NOTIFICATION: - After an IM dose is administered, the doctor on duty will be notified. - For fetus monitoring refer to peri-natal consultant																						
		<table><tr><th>Medication</th><th>Recommended maximum single PRN Dose</th><th>Total maximum oral AND IM daily dose (including regular)</th></tr><tr><td>Oral Quetiapine</td><td>50mg</td><td>400mg</td></tr><tr><td>Oral Lorazepam</td><td>2mg</td><td>4mg</td></tr><tr><td>Oral Olanzapine</td><td>5mg</td><td>30mg</td></tr><tr><td>IM Midazolam</td><td>5mg</td><td>10mg</td></tr><tr><td>IM Lorazepam</td><td>1mg</td><td>4mg</td></tr><tr><td>IM Olanzapine</td><td>5mg</td><td>30mg</td></tr></table>		Medication	Recommended maximum single PRN Dose	Total maximum oral AND IM daily dose (including regular)	Oral Quetiapine	50mg	400mg	Oral Lorazepam	2mg	4mg	Oral Olanzapine	5mg	30mg	IM Midazolam	5mg	10mg	IM Lorazepam	1mg	4mg	IM Olanzapine	5mg	30mg
Medication	Recommended maximum single PRN Dose	Total maximum oral AND IM daily dose (including regular)																						
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